Member 4 Presentation Masterclass

Complete UI Breakdown, Every Navigation Section, Every Button & Slider Explained

ROLE: FRONTEND UI/UX & LIVE DEMO LEAD | **DIFFICULTY: RANK 2 (VISUAL & DEMO-DRIVEN)** | **PRESENTATION DURATION: ~3.0 MINUTES**

Your Core Mission: As Member 4, you are the **Showstopper** of the presentation. While Member 1 explains the problem and Member 2 explains the math, **YOU make it come alive on the screen**. You will click the buttons, explain every visual cue, move the sliders, and show how Quality-Aware routing detours traffic in real time!

1. OVERVIEW OF THE DASHBOARD SCREEN



Figure 1: NetResQ Operations Dashboard showing header controls, 4 metric cards, and interactive SVG topology.

2. EXPLAINING THE HEADER BAR (EVERY CONTROL EXPLAINED)

HEADER 1 **NetResQ Wordmark & Heartbeat Pulse Icon**

What it is: Top-left branding with a glowing pulse icon.

> **What to say:** *"NetResQ stands for Network Rescue — our Layer 3 adaptive self-healing network control system."*

HEADER 2 **'BACKEND LIVE' Badge (Green / Red)**

What it is: Active health polling indicator querying `GET /api/health` every 3 seconds.

> **What to say:** *"This green badge confirms that our FastAPI control plane engine is running live and communicating with the React dashboard via REST API polling."*

HEADER 3 **'SIMULATION MODE' Badge (Purple)**

What it is: Indicates that the system is operating in standalone software emulation mode.

> **What to say:** *"This displays our active operating mode. NetResQ runs standalone in simulation mode without requiring Linux root privileges, but also supports real SDN controllers via Ryu and Mininet."*

HEADER 4 **Route Selectors: [H1] → [H4]**

What it is: Dropdowns that let the operator select source and destination hosts (H1, H2, H3, H4).

> **What to say:** "*These dropdowns allow us to test and inspect end-to-end paths between any pair of hosts across our 3-switch triangle core.*"

HEADER 5 **Algorithm Toggle: [Quality-Aware] vs [Baseline Dijkstra]**

What it is: Toggle switch that changes the active routing algorithm on the live canvas.

> **What to say:** "*This toggle lets us instantly switch the displayed active forwarding route between conventional hop-count Dijkstra and our Quality-Aware multi-metric algorithm.*"

HEADER 6 **Amber 'Demo Congestion' Button 🔥 (YOUR KEY ACTION)**

What it is: One-click trigger that injects 65ms latency, 18% packet loss, and 85% utilization on direct link L_S1_S3.

> **What to say:** "*This is our primary demonstration trigger. Clicking 'Demo Congestion' injects severe delay and packet drops on link L_S1_S3, forcing Quality-Aware routing to dynamically reroute via Switch 2!*"

HEADER 7 **'Reset Scenario' Button**

What it is: Restores all link metrics, status, and routes back to the default healthy state.

> **What to say:** "*Clicking 'Reset Scenario' safely resets the topology back to its clean baseline state.*"

3. THE 4 TELEMETRY METRIC CARDS

Metric Card	Value Shown	What It Measures	What You Say To Evaluator
Switches Online	`3`	Triangle core switches (S1, S2, S3)	"Shows all 3 OpenFlow switches in the triangle core are online and active."
Link Health Status	`7 / 7 Links UP`	3 core links + 4 host edge links	"Monitors physical links. When a link fails, it alerts how many links are DOWN."
Average Link Latency	`5.5 ms`	Network-wide average one-way delay	"Real-time average delay across active links, with packet loss and utilization."
Network Health & Recalc	`100% | 0.48ms`	Health score + Software recalculation time	"Health gauge penalized by loss and delay. Recalc time (0.48ms) is the CPU time required by Dijkstra to reroute traffic."

4. THE INTERACTIVE TOPOLOGY CANVAS & SLIDING LINK INSPECTOR



Figure 2: Clicking any link opens the right-hand Link Inspector panel with real-time sliders for latency, packet loss, and utilization.

4.1 Understanding the Visual Styles on the Canvas:

- **Switches (S1, S2, S3):** Dark rounded cards with blue borders and DPID identifiers.
- **Hosts (H1, H2, H3, H4):** Circular endpoints showing IP addresses (e.g. `10.0.0.1` to `10.0.0.4`).
- **Green Solid Lines:** Healthy active links.
- **Amber Solid Lines:** Degraded links suffering high latency or loss.
- **Red-Dashed Lines:** Failed / severed links (status = `DOWN`).
- **Neon Cyan Glowing Flowing Line:** The **ACTIVE FORWARDING ROUTE**. Animated using CSS keyframe dash-offsets to represent real packets moving across the path!

4.2 Explaining the Sliding Inspector Window (Right Panel):

When you click on any link (e.g., `L_S1_S3`), the right panel slides open:

- **Link ID & Endpoints:** Shows `L_S1_S3 (S1 ↔ S3)`.
- **Fail Link / Restore Link Button:** Manually cuts or restores the link.
- **Latency Slider (1ms - 150ms):** Simulates fiber distance or propagation delays.
- **Packet Loss Slider (0% - 50%):** Simulates packet drops caused by queue overflow.
- **Utilization Slider (0% - 100%):** Simulates traffic bandwidth congestion.
- **'Apply Metrics Update' Button:** Sends `POST /api/links/{id}/metrics` to re-evaluate the graph.

> **What to say:** *"Our interactive inspector allows the operator to select any link and manually test line failures or adjust latency and loss sliders to see the routing engine adapt in real time."*

5. TAB 3: ROUTING ANALYSIS & PERFORMANCE COMPARISON

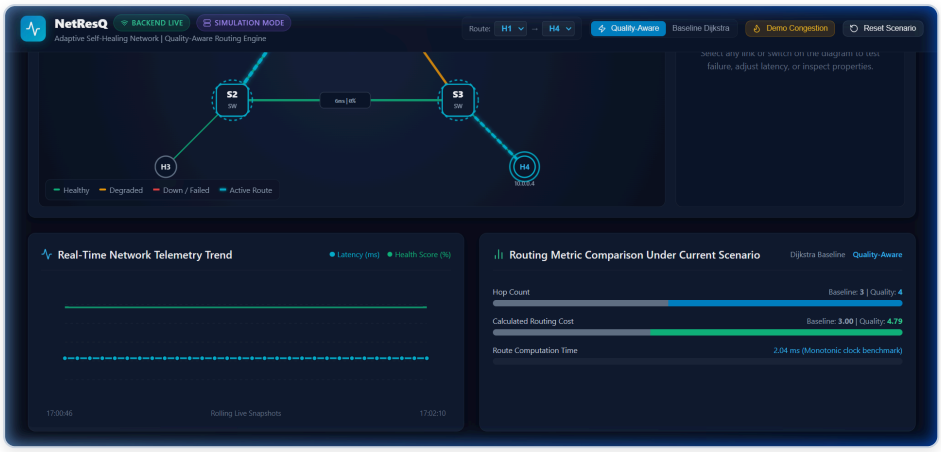


Figure 3: Side-by-side comparison between Baseline Dijkstra and Quality-Aware Routing with the Hop Breakdown Table.

When you click the **"Routing Analysis & Proof"** tab, you show this screen. Here is how to explain every section:

CARD 1 **Baseline Dijkstra Routing (Left Card)**

Shows: H1 → S1 → S3 → H4 (3 hops, Cost: 3.0).

> **What to say:** *"Notice on the left: Baseline Dijkstra only minimizes hop count. Even when link L_S1_S3 is degraded with 18% packet loss, Dijkstra is trapped on the direct 3-hop route because 3 hops is mathematically less than 4 hops."*

CARD 2 **Proposed Quality-Aware Routing (Right Card)**

Shows: H1 → S1 → S2 → S3 → H4 (4 hops, Cost: 4.79 vs 6.12).

> **What to say:** *"On the right, NetResQ's Quality-Aware routing penalizes the lossy link. It automatically chooses the 4-hop path through S2 because its composite cost (4.79) is cheaper than the degraded direct path (6.12)!"*

TABLE **Active Path Hop Breakdown Table**

Lists every link on the chosen route with exact measurements: Latency (ms), Loss (%), Utilization (%), and Calculated Edge Cost.

> **What to say:** *"This table provides full explainability, showing the exact latency and cost breakdown for each individual hop on the selected route."*

6. TAB 4: OPERATIONAL EVENTS SECTION (EVERY ELEMENT EXPLAINED)

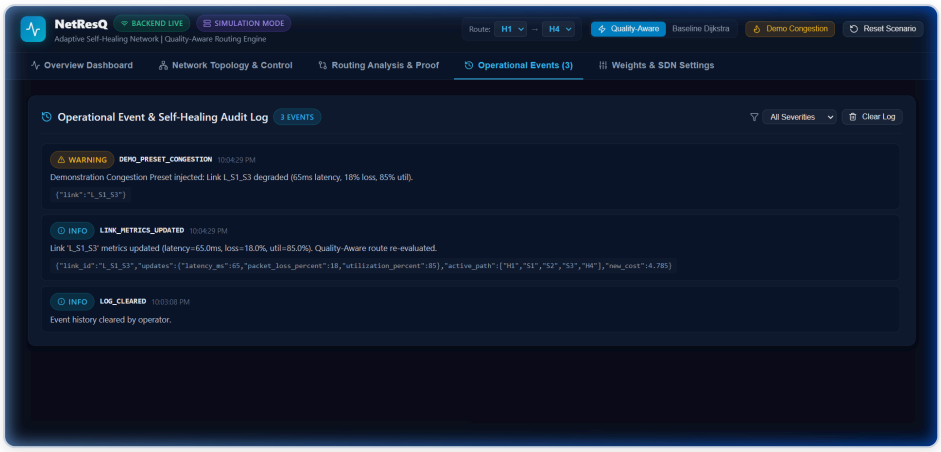


Figure 4: Operational Events Audit Log showing severity filter dropdown, Clear Log button, and chronological event stream.

When you click the **"Operational Events"** tab in the navigation bar, it displays the live audit log of all system transitions. Here is what every control does:

EVENT 1 **Severity Filter Dropdown ('All Severities')**

What it does: Allows the operator to filter the audit log by severity level: `All Severities`, `Errors Only`, `Warnings Only`, `Success Only`, and `Info Only`.

> **What to say:** *"This filter dropdown lets an administrator isolate critical alerts — such as network partition errors or link failure warnings — from routine info messages."*

EVENT 2 **'Clear Log' Button (Trash Icon)**

What it does: Sends a `POST /api/events/clear` request to clear the displayed event history and logs a fresh audit entry stating that the log was cleared by the operator.

> **What to say:** *"Clicking 'Clear Log' purges the audit view for a clean demonstration run without affecting the underlying network state."*

EVENT 3 **Severity Badges (INFO, WARNING, ERROR, SUCCESS)**

What it does: Visual badges color-coding the nature of the network event:

- `INFO`: System startup, metric parameter updates, simulation resets.
- `WARNING`: Link failure where an alternate path was found, demo congestion injected.
- `ERROR`: Complete network partition or unreachable destination.
- `SUCCESS`: Link successfully restored to service.

> **What to say:** *"Our audit engine assigns distinct severity levels: green for restorations, yellow for link failovers, and red for partitioned destinations."*

EVENT 4 **Event Type & Timestamps**

What it does: Displays the event tag (e.g. `LINK_FAILURE`, `LINK_RESTORED`, `SIMULATION_RESET`) alongside local/UTC timestamps.

> **What to say:** *"Each event is stamped with millisecond precision and recorded in reverse chronological order, making it straightforward to trace recovery timelines."*

EVENT 5 **JSON Details Tag**

What it does: Displays a dark code chip at the bottom of the card containing exact metadata, such as the previous path, new path, and calculated failover duration (e.g. `{"recalculation_time_ms": 0.48}`).

> **What to say:** "*This metadata chip provides machine-readable diagnostic details showing both the old path and the newly elected alternate route.*"

7. TAB 5: WEIGHTS & SDN SETTINGS SECTION (EVERY BUTTON & SLIDER EXPLAINED)

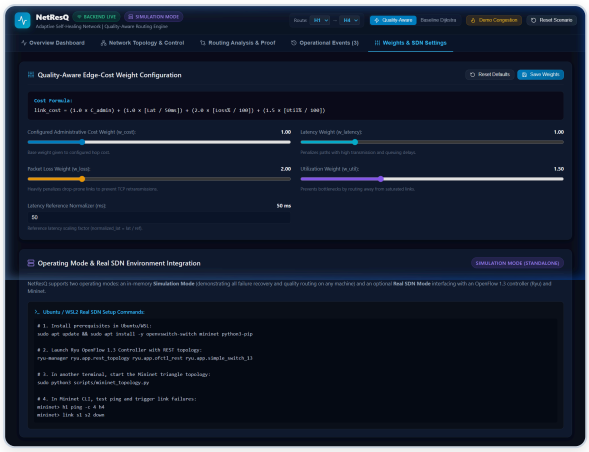


Figure 5: Weights & SDN Settings tab showing live formula box, 5 sliders/inputs, Save/Reset buttons, and Mininet setup instructions.

When you click the **"Weights & SDN Settings"** tab, you have full control over the mathematical routing engine and environment configuration:

TOP BAR **'Reset Defaults' Button**

What it does: Instantly resets all 4 weighting coefficients (1.0, 1.0, 2.0, 1.5) and the 50ms reference latency back to standard defaults and recalculates routes.

> **What to say:** *"Clicking 'Reset Defaults' restores the recommended balance between hop count, latency, packet loss, and utilization."*

TOP BAR **'Save Weights' Button**

What it does: Sends a `POST /api/weights` request to update the active weighting policy on the backend and triggers dynamic path re-evaluation across all host pairs.

> **What to say:** *"Clicking 'Save Weights' commits our custom weights to the backend routing engine, triggering an immediate re-evaluation of all paths."*

MATH BOX **Live Mathematical Cost Formula Box**

What it does: Displays the live, interactive cost formula with the currently configured numerical coefficients:

```
link_cost = (w_cost × C_admin) + (w_latency × [Lat / ref_ms]) + (w_loss × [Loss% / 100]) + (w_util × [Util% / 100])
```

> **What to say:** *"This box shows our normalized composite cost equation updating in real time as the sliders are moved."*

The 5 Sliders & Inputs (What Each Does):

Slider / Input	Default Value	Range	What It Does & What You Say
Configured Cost Weight (w_cost)	`1.00`	0.0 to 5.0	"Controls the importance of administrative hop count. Increasing this makes the algorithm behave more like traditional Dijkstra."
Latency Weight (w_latency)	`1.00`	0.0 to 5.0	"Penalizes propagation and queuing delays. Useful for interactive audio/video traffic."

Slider / Input	Default Value	Range	What It Does & What You Say
Packet Loss Weight (w_loss)	`2.00`	0.0 to 10.0	"Heavily penalizes drop-prone links. We set it to 2.0 because packet loss causes TCP timeouts and throughput collapse."
Utilization Weight (w_util)	`1.50`	0.0 to 5.0	"Penalizes saturated links to avoid buffer overflow and distribute traffic evenly."
Latency Reference Normalizer	`50 ms`	10 to 500 ms	"Normalizes milliseconds into decimals: normalized_lat = latency / ref_ms."

`SDN PANEL` **Operating Mode & Ubuntu/WSL2 Real SDN Setup Box**

What it does: Displays the current operating status badge (`Simulation Mode (Standalone)`) and copy-pasteable terminal commands for launching the **Ryu OpenFlow 1.3 Controller** and **Mininet** on Ubuntu or Windows Subsystem for Linux (WSL2).

> **What to say:** "*This lower panel documents our real-world SDN integration. NetResQ includes ready-to-run Python scripts for launching Mininet with Open vSwitch and the Ryu OpenFlow 1.3 controller under Linux or WSL2.*"

8. THE FORMULA PART: HOW TO EXPLAIN IT SIMPLY

```
normalized_latency = latency_ms / 50.0 ms normalized_loss = packet_loss_percent / 100.0 normalized_util = utilization_percent / 100.0 link_cost = (1.0 × C_admin) + (1.0 × norm_latency) + (2.0 × norm_loss) + (1.5 × norm_util)
```

The 3-Sentence Formula Punchline (Memorize This!):

> **"We normalize latency against 50ms, and packet loss and utilization into decimals between 0 and 1. We assign the highest weight (2.0) to packet loss because packet drops destroy TCP throughput. When direct link L_S1_S3 suffers 18% loss, its calculated cost spikes from 1.44 to 3.94, making the 4-hop clean route via S2 mathematically cheaper (4.79 vs 6.12)!"**

9. STEP-BY-STEP LIVE DEMO SCRIPT (WORD-FOR-WORD WALKTHROUGH)

Step 1 — Introduce the UI (30 Seconds):
"Respected evaluators, I am [Name], and I will demonstrate the NetResQ Operations Dashboard. As you can see, the UI is designed like a carrier-grade Network Operations Center. In our 3-switch triangle core, all 7 links are currently green and healthy. The active route between Host H1 and Host H4 is highlighted by this glowing cyan animated line, representing live packets flowing across the direct 3-hop route."

Step 2 — Click 'Demo Congestion' (45 Seconds):
"Now, let me trigger our primary test case. I will click the amber 'Demo Congestion' button in the header. Notice what happens immediately: direct link L_S1_S3 turns yellow with 65ms latency and 18% packet loss. Watch the glowing cyan active route: it has automatically detoured through Switch 2! Even though going through S2 takes 4 hops instead of 3, our Quality-Aware algorithm avoids the congested link to protect user quality."

Step 3 — Show the Comparison Proof (45 Seconds):
"Let us click on the 'Routing Analysis & Proof' tab. Look at the two cards side-by-side: Baseline Dijkstra is trapped on the direct 3-hop link with 18% packet loss. But NetResQ's Quality-Aware routing chose the 4-hop path because its composite cost is only 4.79 compared to 6.12. The table below shows the exact latency and cost breakdown for each individual hop."

Step 4 — Show Link Failure & Self-Healing (45 Seconds):
"Now back on the topology, let me simulate a total fiber cut. I click link L_S1_S3 and click 'Fail Link'. The link turns red-dashed. Our self-healing engine detects the break and recalculates an alternate route in just 0.48 milliseconds, as audited in our Operational Events log. When I click 'Restore Link', the link turns green and normal service resumes."

Step 5 — Show Weights & Conclude (30 Seconds):
"Finally, in our Weights & SDN Settings tab, we can adjust policy sliders in real time or reset to defaults. Clicking 'Reset Scenario' returns our network back to its clean baseline state. This completes our live demonstration of NetResQ."